Christol's Theorem, Automata, and the Pandrosion Frobenius

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Abstract

We connect Christol's theorem—a formal power series $f \in \mathbb{F}_p[[x]]$ is algebraic over $\mathbb{F}_p(x)$ iff its coefficient sequence is p-automatic—to the Pandrosion framework through the Frobenius endomorphism $P(x)^p = P(x^p)$ over $\mathbb{F}_p$. The Cartier operators C_r, which extract the r-th subsequence in base p, are interpreted as Pandrosion fiber extractions. We derive and verify the algebraic equation $f^2(1+x)^3 + f(1+x)^2 + x = 0$ over $\mathbb{F}_2$ for the Thue–Morse series, compute p-kernels for three test sequences (Thue–Morse: 2 states, Baum–Sweet: 3 states, Sturmian: growing), and confirm that the Frobenius kills the Pandrosion field: $(P^p)' = 0$ over $\mathbb{F}_p$.

1 Christol's theorem

Theorem 1.1 (Christol, 1979)**.** *A formal power series $f(x) = \sum_{n \geq 0} a_n x^n \in \mathbb{F}_p[[x]]$ is algebraic over $\mathbb{F}_p(x)$ if and only if the sequence $(a_n)_{n\geq 0}$ is p-automatic.*

Definition 1.2. A sequence $(a_n) \in \mathbb{F}_p^{\mathbb{N}}$ is *p-automatic* if its *p-kernel*

$$\mathcal{K}_p = \left\{ \left(a_{p^k n + r}\right)_{n \geq 0} : k \geq 0,\ 0 \leq r < p^k \right\} \tag{1}$$

is finite. Equivalently: (a_n) is generated by a finite automaton reading the base-p digits of n.

2 The Pandrosion Frobenius

Theorem 2.1 (Frobenius identity)**.** *Over $\mathbb{F}_p$ for any polynomial P:*

$$\boxed{P(x)^p = P(x^p).} \tag{2}$$

Proof. $P(x)^p = \left(\sum c_k x^k\right)^p = \sum c_k^p x^{pk} = \sum c_k x^{pk} = P(x^p)$, using $c_k^p = c_k$ in $\mathbb{F}_p$. □

Corollary 2.2 (Frobenius kills the derivative)**.** $(P^p)' = p\, P^{p-1} P' = 0$ *over $\mathbb{F}_p$. In Pandrosion language: $F_{P^p} = (P^p)'/P^p = 0$. The Pandrosion field of a p-th power is **trivial**.*

Verified for $p = 2, 3, 5, 7$: the derivative of P^p is identically zero modulo p.

3 The Thue–Morse series

Definition 3.1. The Thue–Morse sequence $t(n)$ is defined by $t(0) = 0$, $t(2n) = t(n)$, $t(2n+1) = 1 - t(n)$. Equivalently, $t(n) = s_2(n) \bmod 2$ (digit sum in base 2).

Theorem 3.2 (Algebraicity of Thue–Morse)**.** *The series $f(x) = \sum t(n)\, x^n \in \mathbb{F}_2[[x]]$ satisfies:*

$$\boxed{f^2(1+x)^3 + f(1+x)^2 + x = 0 \qquad \textit{over } \mathbb{F}_2.} \tag{3}$$

Proof. Over $\mathbb{F}_2$: $f(x^2) = f(x)^2$ (Frobenius, Theorem 2.1). The recurrence gives:

$$f(x) = f(x^2) + x\left(\sum_{n\geq 0} x^{2n} + f(x^2)\right) = f^2(1+x) + \tfrac{x}{1+x^2}. \tag{4}$$

Multiplying by $(1+x^2) = (1+x)^2$ over $\mathbb{F}_2$: $f(1+x)^2 = f^2(1+x)^3 + x$. □

Verified numerically: both the functional equation and the polynomial equation hold with 0 errors modulo x^{128}.

4 Cartier operators as Pandrosion fibers

Definition 4.1. The *Cartier operator* C_r $(0 \leq r < p)$ extracts the r-th subsequence:

$$C_r\left(\sum a_n x^n\right) = \sum a_{pn+r}\, x^n. \tag{5}$$

Proposition 4.2 (Thue–Morse Cartier)**.** *$C_0(f) = f$ and $C_1(f) = 1 - f$. The orbit $\{C_{r_1} \circ \cdots \circ C_{r_k}(f)\} = \{f, 1-f\}$ has 2 elements.*

This is the *Pandrosion fiber decomposition in base p*: the Cartier operator extracts the "r-th fiber" of the coefficient sequence, and finiteness of the orbit is exactly the automaticity criterion.

5 p-kernel verification

Sequence	$\lvert\mathcal{K}_2\rvert$	**Automatic?**	**Algebraic over $\mathbb{F}_2$?**
Thue–Morse	2	Yes	Yes: $f^2(1+x)^3 + f(1+x)^2 + x = 0$
Baum–Sweet	3	Yes	Yes (degree 3 equation)
Sturmian($\sqrt{2}$)	46+ (grows)	No	No (irrational slope)

The kernel sizes distinguish sharply: automatic sequences have $\lvert\mathcal{K}_2\rvert \in \{2, 3\}$, while the Sturmian sequence (irrational rotation) has a kernel that grows with depth—confirming non-automaticity and hence non-algebraicity over $\mathbb{F}_2$.

6 Connection to Pandrosion dynamics

Christol ingredient	**Pandrosion tool**	**Paper**
Frobenius $f(x^p) = f(x)^p$	$P^p = P(x^p)$, $(P^p)' = 0$	71
p-kernel finiteness	Cartier orbit of fibers	This paper
Power sums mod p	F_P Laurent coefficients	70
$x^p - x = \prod(x-a)$	Fermat–Pandrosion	71
Algebraic equations	Polynomial arithmetic over $\mathbb{F}_p$	71, 73

References

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